

HW Starter — Agentic RAG

This is your code skeleton.

Read in this order:

- 1. `./HW_INSTRUCTIONS.md` — assignment writeup, prerequisites, setup, submission, rubric
- 2. `MILESTONES.md` — the build checklist (M1 → M7) with hint-level guidance and the inline pre-fill walkthrough

Run the agent (after `uv sync`):

```
uv run python main.py
```

The starter ships with `TEST_TO_RUN = "ingestion"` set in `main.py`. Your first run will hit a `NotImplementedError` pointing you at M3 — that's expected.

File roles

File	Status
<code>prompts.py</code>	Provided. Read in M1.
<code>config.py</code>	Provided. The <code>CONFIG</code> dict has all hyperparameters. Read in M1.
<code>pyproject.toml</code>	Provided. <code>uv sync</code> generates <code>uv.lock</code> .
<code>env.template</code>	Provided. Copy to <code>.env</code> and add your four API keys.
<code>ingestion.py</code>	TODO body in <code>load_and_process_filings</code> . Built in M3.
<code>retriever.py</code>	TODO bodies in <code>create_vector_store</code> and <code>create_retriever_with_reranking</code> . Built in M4.
<code>tools.py</code>	TODO bodies in three <code>_run</code> methods plus two helpers on <code>CompanyDirectorsTool</code> . Built in M5.
<code>main.py</code>	TODO bodies in <code>build_pipeline</code> , <code>build_tools</code> , <code>build_agent</code> , and the M7 invocation block. Built in M6 + M7.

Test runner switch

In `main.py`, change `TEST_TO_RUN` to verify each milestone independently:

```
TEST_TO_RUN = "ingestion"    # M3 done?
TEST_TO_RUN = "retriever"    # M4 done?
TEST_TO_RUN = "tools"        # M5 done?
TEST_TO_RUN = None           # M7 - full agent end-to-end
```

The directors tool burns ~8 SerpAPI calls per run (one per Tesla director). See the API quota strategy in `HW_INSTRUCTIONS.md`.